Artificial Intelligence CS 401

Clustering – II
K-medoids

Lecture No. 18

National University of Computer and Emerging Sciences,
Lahore

CLUSTER ANALYSIS

Partitioning Methods: k-medoids

Instead of mean we nominate a median of the cluster
This makes it less sensitive to outliers

Input: The number of clusters k, and a database containing n objects.

Output: A set of k clusters which minimizes the sum of the dissimilarities of all the objects to their nearest medoid.

Method:

```
arbitrarily choose k objects as the initial medoids;
repeat
        assign each remaining object to the cluster with the nearest medoid;
        randomly select a non-medoid object, o_random;
        compute the total cost, S, of swapping o_j with o_random;
        if S < 0 then swap o_j with o_random to form the new set of k medoids;
until no change;
```

It is computationally expensive

How to find Mediods:

For odd sum:

[(1, 5), (3, 8), (4, 9), (6, 10), (9, 12)]

X = 1, 3, 4, 6. 9

Sum is 5 (odd)

Y = 5, 8, 9, 10, 12

Mediods = Median of each attribute = (4, 9)

For even sum:

X = 2, 4, 5, 7, 9, 10

Y = 3, 4, 6, 8, 12, 14

Sum = 6 (even)

Mediods = mean of center two elements = (6, 7)

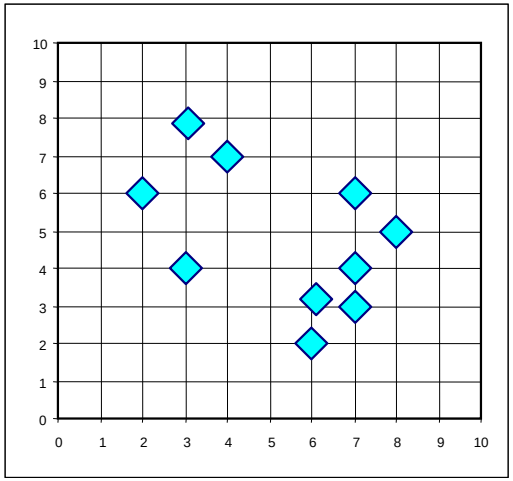
X = (5+7)/2 = 6

Y = (6+8)/2 = 7

CLUSTER ANALYSIS

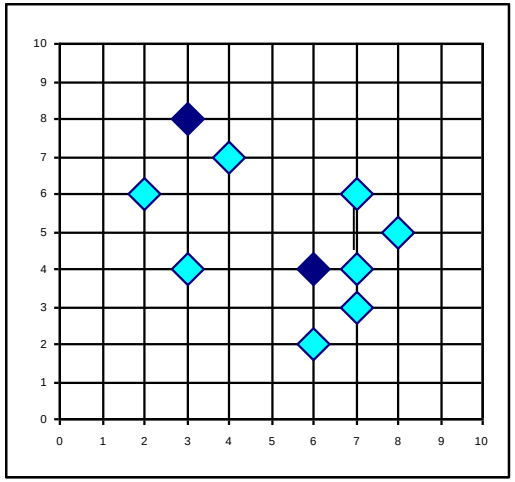
Partitioning Methods: k-mediods & PAM

Total Cost = 20

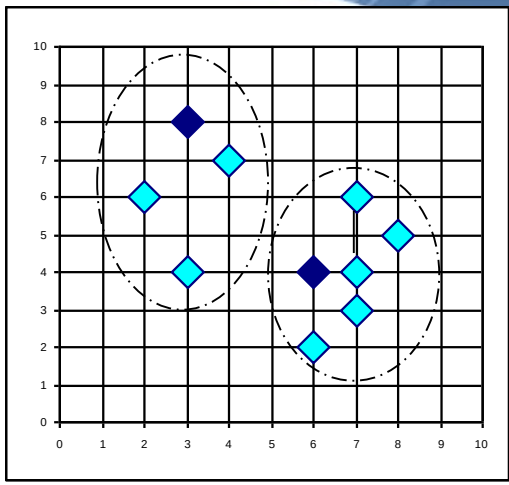


K=2

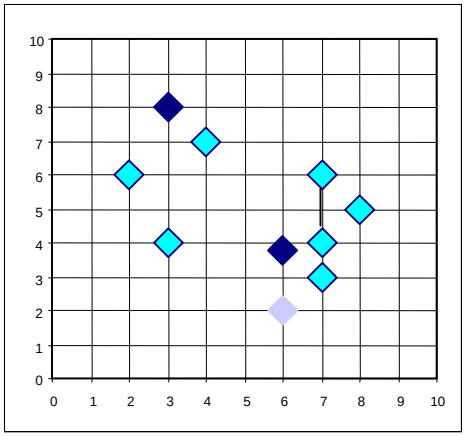
Arbitrary choose k object as initial medoids



Assign each remaining object to nearest medoids

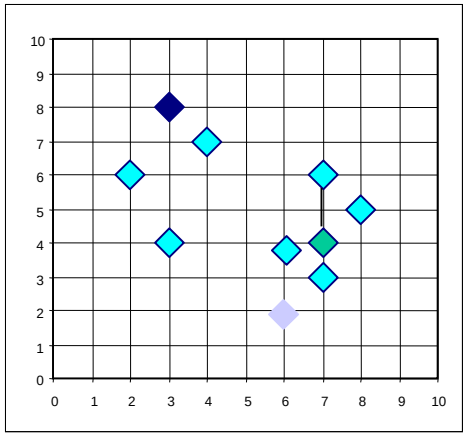


Randomly select a nonmedoid object, O_{ramdom}



Compute total cost of swapping

Total Cost = 26



Swapping O and O_{ramdom}

If quality is improved.

Do loop

Until no change

CLUSTER ANALYSIS

Partitioning Methods: k-means & k-medoids

The mean or median can be used as a model for its cluster

A new example can be assigned to the cluster on the basis of smallest distance between the cluster mean (median) and the example

Thus the data space is *partitioned* into *k* regions

A medoid can be defined as the object of a cluster whose average dissimilarity to all the objects in the cluster is minimal, that is, it is a most centrally located point in the cluster.

CLUSTER ANALYSIS

Partitioning Methods

Construct a partition of a database *D* of *n* objects into a set of *k* clusters. Given a *k*, find a partition of *k clusters* that optimizes the chosen partitioning criterion

Global optimal: exhaustively enumerate all partitions

Heuristic methods: *k-means* and *k-medoids* algorithms

k-means **(MacQueen'67): Each cluster is represented by the center of the cluster**

k-medoids **or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster**

CLUSTER ANALYSIS

Partitioning Methods: CLARA

(Clustering Large Applications)

Proposed in 1990

Implementation of the Partitioning Around Mediods algorithm in a subset of the dataset.

It draws multiple samples of the dataset, applies PAM on the samples, and then outputs the best clustering out of these samples.

CLUSTER ANALYSIS

Partitioning Methods: CLARANS

(Clustering Large Applications based upon Randomized Search)

It usually starts with an arbitrary set of *k* medoids

It picks a medoid and replaces it by a non-medoid. If it is more suitable as a medoid, it is made a medoid

If it is not a better medoid, a counter is incremented

If the value of the counter reaches a pre-set number, a local minimum is said to be found and algorithm restarts with a new random set of medoids

This process is repeated n times and the best set of medoids is retained

CLUSTER ANALYSIS

Reading Assignment

Chapter 8 of Data Mining by Hans & Kamber

Q1. Implement K-Medoids Algorithm on the data given in Figure 1. The value of k is 2. (20)

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Figure 1

Class Activity

Q1. Implement k-means algorithm on the data given in Figure 1. The value of k is 3. Use Euclidean distance as the distance measure. (20)

$$d(\mathbf{p},\mathbf{q}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2}.$$

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Figure 1